

Plan B – Backup Scenarios for Scallop Research Track Assessment:

CASA -> SYM -> Reference Points

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Term of Reference 8

Atlantic Sea Scallop Research Track

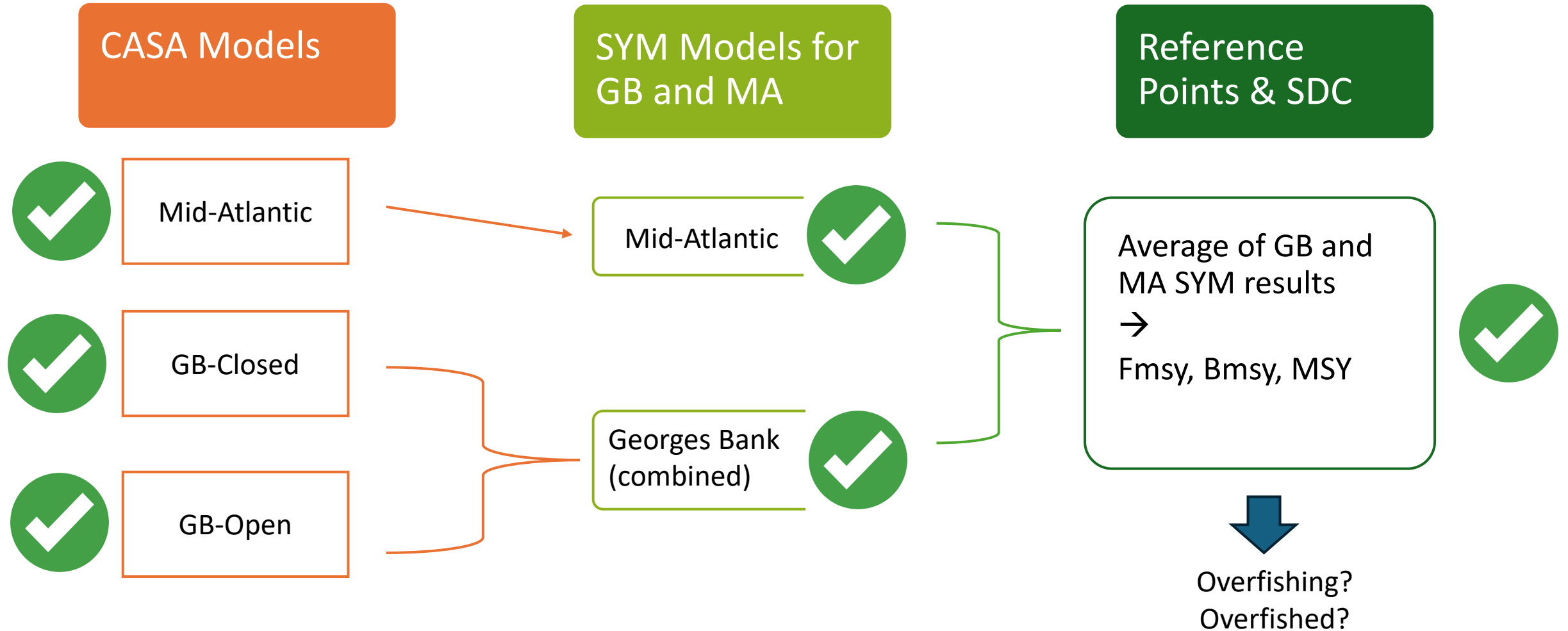
Term of Reference 8

Develop a backup assessment approach for providing scientific advice to managers if the proposed assessment approach does not pass peer review or the approved approach is rejected in a future management track assessment.

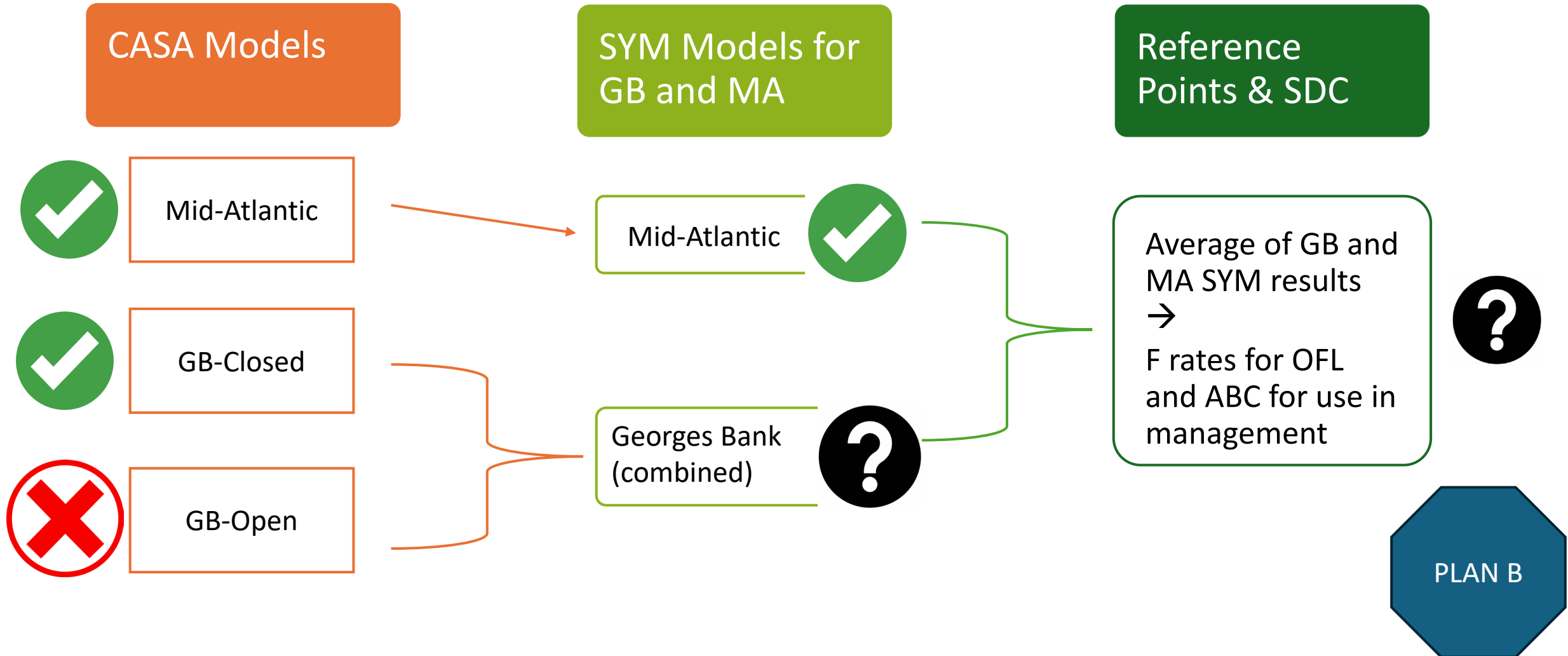
Steps where current assessment might fail:

1. The Catch at Size Analysis model (CASA) fails to perform for any of the three management areas.
 - *The CASA model estimates the number of fully recruited scallops in the population for each management region (Mid-Atlantic, Georges Bank closed, and Georges Bank open).*
2. The Stochastic Yield Model (SYM) fails to perform for any of the two areas (Mid-Atlantic or the combined Georges Bank area).
 - *The SYM combines the Mid-Atlantic and Georges Bank results to derive a stock-wide result. Using a Monte Carlo simulation, candidate reference points are provided, including recommendations for biological reference points of FMSY, BMSY, and MSY. SYM accounts for uncertainty in the CASA model estimates and addresses the additional issues of unaccounted mortality during the fishing process (e.g. bycatch mortality of discards and incidental mortality caused by gear on bottom). The Fmsy from the SYM model is used in status determination.*

Steps for development of Reference Points



Scenarios – CASA Model Fails (Hypothetical)



Three scenarios considered:

- CASA Backup
- SYM Backup
- Model Free Backup



Wikipedia – Dann Blackwood – USGS

Proposed CASA Backup

- The scallop dredge surveys are designed in such a way that the number of scallops observed by area of bottom covered by tow and corrected for catchability can provide a measure of average scallop density by area. This density by area can be expanded by strata (i.e., SAMS area) to get the number of scallops by strata for each SAMS management area. These population estimates can be fed into the SYM to calculate reference points.
- Similar calculations can be made using optical and other survey methods (See TOR 3 working paper).

Proposed SYM Backup

- The SYM was outlined in an ICES peer reviewed journal article (Hart 2013) and has not, in substance, changed since its use in 2014 or during the 2018 Stock Assessment Workshop (65th SAW), when it was approved for use. The research track working group is recommending changes to M and recruitment in SYM in this assessment.
- Reference points can be calculated deterministically using means or medians or trimmed means of the various CASA output parameters rather than through stochastic simulation. However, it is known that stochastic estimates of $FMSY$ are lower than deterministic estimates as only the stochastic reference points take into account the risk that $FMSY$ has been overestimated due to parameter misestimation. So conservative use of such deterministic reference points is recommended.

Proposed Model Free Backup

- Should the failure of approval for any or all of the model steps in Steps 1 and 2 occur in such a way that status determination remains unknown there are other options that can be considered. For example, the scallop dredge surveys or the optical surveys could still be used to determine standing stock biomass. In this scenario, a conservative determination of fishing mortality would be determined by using the realized catch over the last five years divided by area swept abundance estimates to come up with an estimate of the fishing mortality reference point proxy for the fishery. However, under such a scenario the productivity of the stock may still remain unknown and consequently certain status determination metrics such as the Overfishing Limit (OFL) would remain unavailable.

Conclusions and WG Recommendations

- The working group recommends that if a backup plan is needed, the adjusted area survey measures (e.g., dredge, drop camera, HabCam) of fully recruited scallop density expanded by survey strata (i.e., SAMS areas), averaged appropriately and summed over strata can be used to arrive at an estimate of scallop abundance.
- This abundance estimate can be combined with similar estimates for the other management region or with the CASA estimates from each region to be processed by the SYM to arrive at the appropriate reference point recommendations.
- If the SYM should fail, then deterministic calculations using means, medians or trimmed means should replace the stochastic calculations.
- Should model steps prove to be inadequate, then a model-free approach should be used to determine exploitation reference points using area swept estimates divided by time averaged landings.